

Scientific writing Swiss TPH: Lesson 1: Introduction to scientific writing and peer reviewing

Lecturer: Dr Riana Paola

This script is to be used together with the lesson slides.



A journey of a thousand miles begins with a single step. (Lao Tzu)

Contents

1. Course objectives	1
2. Academic genres and style	2
2.1 Linguistic features of academic style	3
3. The structure of a thesis and/or research article	6
3.1 The Academic Phrasebank	7
4. Peer reviewing	8
4.1 Peer reviews in scientific journals and in this course	8
4.2 Feedback grid for scientific writing	Error! Bookmark not defined.
4.3 The assignment review process in this course	14
Assignment 1	7

1. Course objectives

This course aims to:

- Support participants in drafting a manuscript such as a research article or thesis.
- Use authentic materials to examine and analyse as examples of scientific text production.
- Guide participants in searching for, identifying and applying academic writing practices.
- Guide participants in drafting and refining their own manuscripts.
- Practise writing, peer reviewing and revising.
- Focus on the structure of an article, a thesis, and a research proposal.
- Provide guidance on and address problems with English language use in academic settings.
- Help participants to learn to write effectively, concisely, and clearly.

1.1 Objectives Lesson 1

- Introduce you to the course, assignments and peer reviewing.
- Learn about the genre and features of scientific writing.
- Introduce you to some sources to support your scientific writing.
- Introduce you to reviewing in general and peer reviewing in this course.

2. Academic genres and style

Theses (plural form of 'thesis') and research articles are two types of academic text. We often refer to a text type as a **genre**.

- Identify at least 3 more academic text types.
- What characteristics do they share that make you group them together as 'academic'?

Texts from the same genres are usually similar in structure, content and style. Closely related to genre is the **audience** – the intended readers of the text – and the text **purpose**, that is, the main reason for writing the text.

Discuss the following questions with your fellow students.

- Who are the intended readers of the academic texts you mentioned above?
- Who are the intended readers of your thesis?
- What is the purpose of the texts you mentioned?

You might agree that your immediate audience is your **supervisor and peers**, but the primary audience/readers of your thesis are the **scholarly community** – they set the standards and are critical. Consider these questions:

1. Who forms part of this scholarly community?
2. What are their expectations? Inform yourself by reading publications in your field, attending seminars, etc.
3. How do you think this audience expects you to **structure** your writing?
4. What **type of language and style** does your audience expect you to use?

Now examine the **texts** from your field of study.

1. How are they **structured**?
2. What are the **characteristics** of the language used in these texts? (Search for examples of formality, vocabulary, voice/distance, etc.)
3. How do the structure and language style differ between your text and the texts of students from **other fields**? Discuss this question with your fellow students and make a table of similarities and differences.

Read the following **quotation about academic writing** and discuss the questions:

'The knack for all students regardless of discipline is to pinpoint what is required and model your work accordingly.' (Burnham, 1994:33, in Murray, 2002:96)

1. Do you agree with this statement? Why?
2. To what extent is an academic writer free to deviate from the 'requirements' or 'models' prescribed or used in the scholarly community?
3. Discuss with your fellow students the possible advantages and disadvantages of adhering to or deviating from the expectations of the scholarly community.

For your thesis or article to be accepted for a higher degree at the university or for publication, it is essential that you adhere to the university's **code and guidelines**. There might also be international as well as national guidelines, procedures and principles laid down by professional and quality assurance bodies and, of course, your department and research supervisor (Murray, 2002:41)! Make yourself well aware of these ground rules and work within them; your supervisor should be able to advise you.

2.1 Linguistic features of academic style

The linguistic characteristics of academic style vary among disciplines, but they tend to include the features listed below (Swales and Feak, 2012:14-27). In the boxes, write down examples of these features that you found in the articles or theses you brought with you. If your texts do not contain these features, how do they differ from the examples given?

- **A vocabulary shift in verbs**, for example the preference for single-word nouns of Latinate origin rather than two/three-word phrasal verbs, such as 'tolerate' rather than 'put up with'. Write down some of the verbs in the academic texts you have. Are any contractions used or are the verbs written in full ('does not' or 'doesn't')?

- **A vocabulary shift in nouns**, with a tendency to use academic terminology and long, complex noun phrases such as 'The emergence of English as the international language of communication has been widely documented.' Write down some of the noun phrases in the texts you brought with you.

- **Mostly objective style**: Can you find examples of the use of first-person singular (I) or first-person plural (we)? If they are used, in which parts of the thesis or article?

- **Indirect rather than direct questions** ('Several studies have investigated why some participants comply with the regulations while others do not.' rather than 'Why do some participants comply with the regulations while others do not?')

- **Use of both active and passive voice in parts** of the paper/thesis. Which tense dominates? When is the passive voice used?

- **Condensed/compact style** that is difficult for inexperienced readers but efficient for expert readers who can extract a lot of information from relatively short, compact/condensed texts. Consider the following example of a longer, more informal sentence compared to a compact, more formal sentence, before writing down examples of such a condensed style found in the theses you brought:

Because the technology has improved, it's less risky than it used to be when you install them at the same time, and it doesn't cost so much either.	Improvements in technology have reduced the risks and high costs associated with simultaneous installation.
---	---

- Focus on **capitalization** in the headings. Is the capitalization done in the same way in all the theses or articles? Compare with your fellow students. Are there trends in certain disciplines, or in British versus American English? Consult these websites and compare the way capitalization is done. <http://www.theguardian.com/uk> (British) and <http://www.nytimes.com/> (American). Can you see trends in the use of capitalization?
- **Formality – the use of academic terminology:** Find examples of such terminology. Is any informal language used? (colloquial words, phrasal verbs (e.g. 'come up with' and abbreviated forms. (See exercises on the next page.)

Formal and informal language use exercises

Examine these sentences and paragraphs and **highlight** the instances of informal language use. Thereafter, rewrite the sentences in a more formal way, keeping in mind the characteristics of academic language outline above.

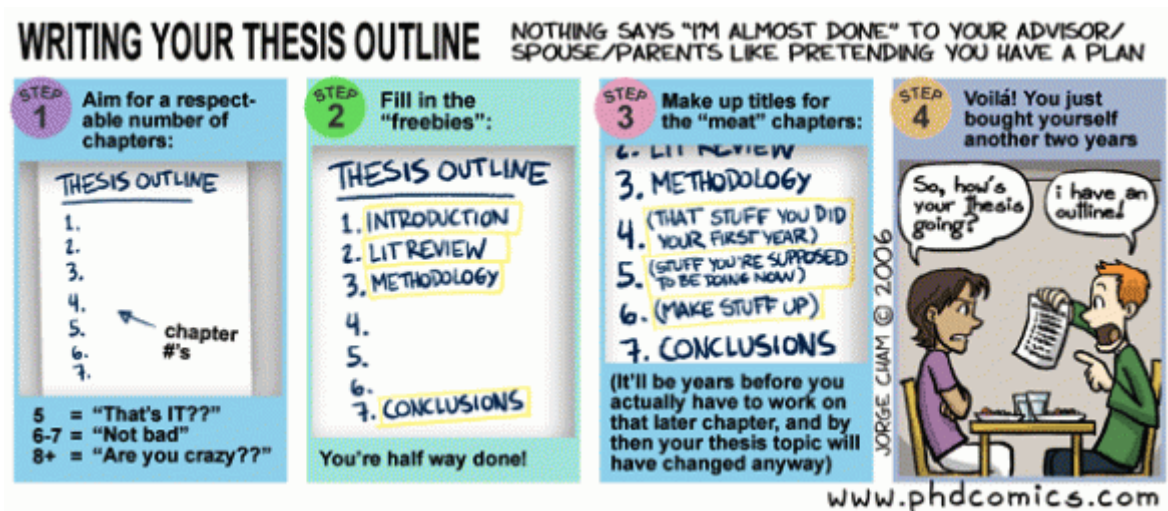
1. Up to now people haven't really looked much into the relationship between mobile phone use and the amount of sport people do.

2. You can use this model to make recycling of plastic more efficient.

3. So why did the financial crisis occur? There are many reasons.

4. There are several studies in epidemiology that have shown that when people don't drink too much, they have a lower risk of developing heart disease in comparison to those people who drink a lot of alcohol (Adapted from Swales and Feak 2012:26).

3. The structure of a thesis and/or research article



[This Photo](#) by Unknown Author is licensed under [CC BY-NC-ND](#)

Discuss these questions:

- What is the usual structure of a research article or thesis in your field of study? What structure do readers in your field of study expect it to have?

Now examine the research article or table of contents of the thesis that you have accessed or brought with you.

- What are the main sections of the article or thesis?
- Are these sections the same as the ones you listed above? How do they differ?
- Are these sections the same as those in articles or theses in other fields of study? Compare with your fellow students.

The generic research article or thesis structure usually looks like this:

1. Introduction/Background/Literature Review
2. Theory/Approach/Methods and Materials/Subjects
3. Results/Analysis
4. Interpretation/Discussion
5. Conclusions/Implications/Recommendations

Murray (2002:116-7, adapted here) suggests using the following **basic outline** when you write a thesis, making sure you include all the points:

1. Introduction /Background /Literature Review (in this unit)
 - Summarize and evaluate books, articles, theses, etc.
 - Define the gap /Establish the niche in the literature that justifies your research.
 - Formulation of hypothesis /Research questions
 - Define and justify your project (filling the gap/niche).
2. Theory /Approach /Methods and Materials /Subjects (See unit 2)
 - Define the methods, theoretical approach, instrument(s).
 - Method of inquiry
 - Show links between your method and others.
 - Justify your method.
3. Results /Analysis (See unit 2)
 - Report what you did, list the steps followed.
 - Document the analysis, showing how you carried it out.
 - Report what you found.
 - Prioritize sections for the thesis or for an appendix.
4. Interpretation /Discussion
 - Interpret what you found.
 - Justify your interpretation.
 - Synthesize results in illustrations, tables, graphs, etc.
5. Conclusions /Implications /Recommendations
 - For future research
 - For future practice
 - Report issues that were beyond the scope of the study.

When writing a research article, much of the same information is included, but in a more concise way.

In the *Swiss TPH Student Handbook* there is a clear outline of what the Swiss TPH expects students to include in each part of their manuscripts.

Discuss these questions:

- What section of your thesis or article are you writing first? Why?
- Do you think there could be other starting points? Which section might be the easiest for you to start with?

In the natural sciences, the Methods and Materials section is often written first, as researchers mostly already have a research or laboratory protocol that they can use as basis. However, one could also start with the introduction and literature review, as they give the general background to the study.

3.1 The Academic Phrasebank

The phrase bank at <http://www.phrasebank.manchester.ac.uk/> , compiled by the University of Manchester, is an excellent resource for academic writing.

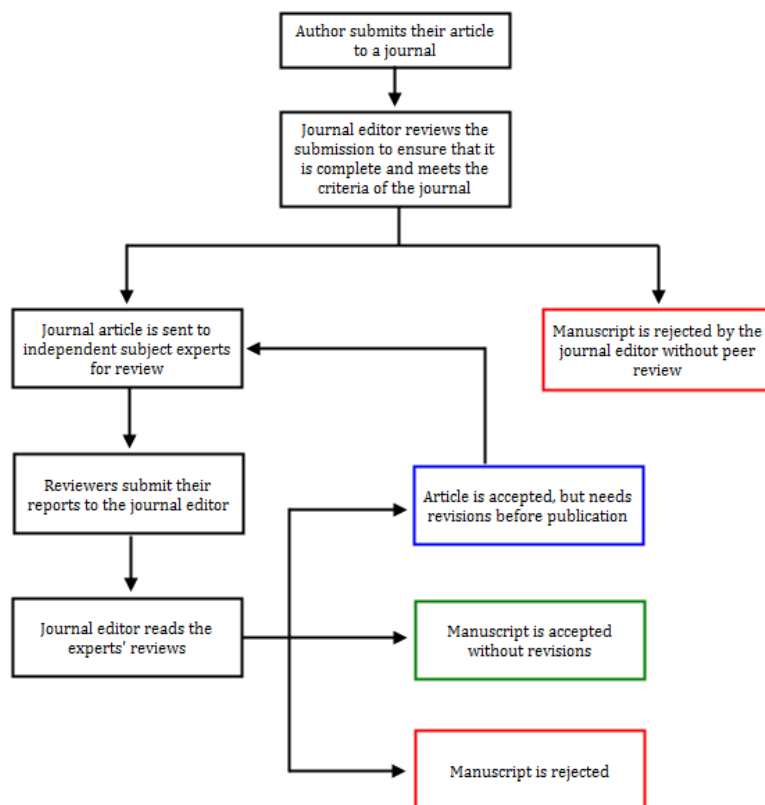
4. Peer reviewing

It is essential that you **fully participate in the peer review process** in this course, as it will help you look critically at how a scientific text is written, at the problems you detect in your review partner's text, and then at the problems in your own texts. In this way, you **learn to look at your own writing from the perspective of the reader, which is a crucial skill that you need to acquire in order to write clearly, cohesively and coherently.**

Uploading your assignments on time is essential, whether it is the first, reviewed or final draft, else you delay the process for the review partners as well as the lecturer, who needs to access the final document before the due date. **Peer reviewing forms part of the course requirements.** See further details below.

4.1 Peer reviews in scientific journals and in this course

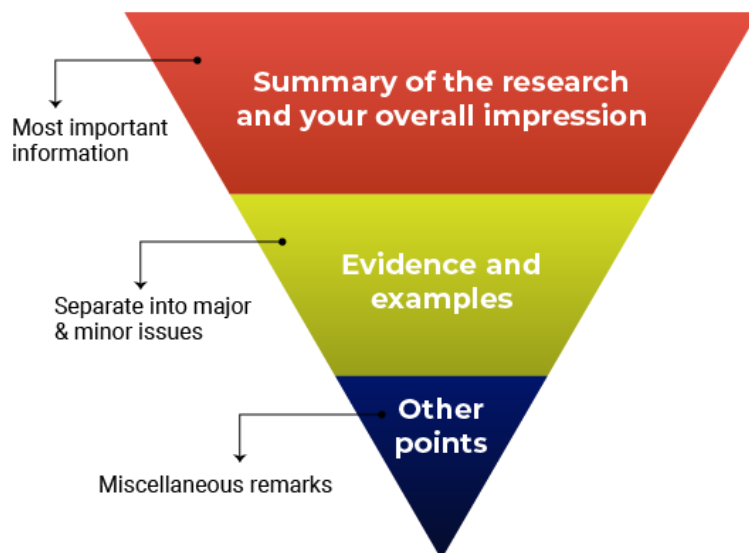
Peer reviews are part of the research and publication process. When writers submit an article for publication, it is sent to reputable reviewers to see if the standard is adequate for the publication, if suitable and recognized methods were used, if the results were interpreted correctly, and if logical conclusions based on the results were drawn. The following infograph summarises the **peer review process** followed by journals:



Discuss these questions about peer reviews in this course with your fellow students:

1. What do you think should be the aims and focus of your peer reviews of assignments? What should you look for when reviewing?
2. How would you describe the attitude of the reviewer?

The following graphic presents an outline of a review of a research paper (<https://plos.org/resource/how-to-write-a-peer-review/>)



We are all rather experienced readers. If you find a paragraph elegant and clear, other readers likely will too. If you find a sentence confusing or wordy, other readers likely will too. You do not need to be an authority on the paper's topic to recognize when the author is communicating information clearly and convincingly.

So, what should you look for when reviewing a peer's writing? Peer reviewing is not only proofreading: it should aim at assisting revision. Not all strengths and weaknesses in texts are equally serious. It can be helpful to use a system of Higher Order Concerns (HOCs) and Lower Order Concerns (LOCs) to plan your feedback strategy (Cf. https://twp.duke.edu/sites/twp.duke.edu/files/file-attachments/providing_feedback%281%29.pdf). For instance, there is little point in highlighting all the comma errors in a paper if the author's central claim (thesis) is not clear. Furthermore, revising for HOCs almost always affects LOCs. When an author revises a thesis statement, for example, s/he will likely also make changes to a paper's organization, tweaking transitions and sentence structures in the process.

When we review each other's texts in this course, we generally only consider a section of a paper or thesis. Use the following checklist for Higher Order, Middle Order and Lower Order Concerns when peer reviewing. The checklist focuses on a full academic article or thesis, so certain points might not apply to each section of a paper.

Higher Order Concerns and Lower Order Concerns checklist for peer review

Higher Order Concerns

Purpose

- Does the manuscript do what the assignment asks it to do?

Thesis

- Does the draft have an explicit thesis statement that focuses and organizes the paper?
- Does the thesis statement make a claim (vs. just stating a non-contestable fact)?

Development

- Is the thesis supported with sound reasoning and solid evidence?
- Are there places in the draft where more details, examples, or specifics are needed?
- Do any paragraphs seem too short or in need of more material?

Organization

- Is there an effective introduction? A satisfying conclusion?
- Do the paragraphs have a clear focus?
- Is there a clear logic to the flow of ideas? Do the transitions work? Is the essay easy to navigate?
- Is the order in which information is presented effective?

Sources: Incorporating the ideas of others

- Are outside sources appropriate for the purpose?
- Are in-text quotations properly introduced and incorporated into the paper's argument?
- Is everything cited that needs to be? Does the author provide citations, including page numbers, for all direct quotes and paraphrases?

A Middle Order Concern: Style

If prose communicates ideas clearly, but lacks flair, style becomes a LOC. When writing style prevents an author from communicating effectively, it becomes a HOC.

- Are sentences clear and easy to follow?
- Is word choice engaging and appropriate?
- Does the language hold your attention?

Lower Order Concerns: Grammar, punctuation, and spelling

Ideally, grammar, punctuation, and spelling are LOCs: a missing comma here or there, an occasional subject-verb disagreement, and a few typos won't make or break a paper. However, when problems with mechanics obscure an author's ability to communicate effectively, they become HOCs.

- Does the author correctly handle mechanical issues such as grammar, punctuation and spelling?
- Are there any errors that detract from your ability to understand what the author is trying to say?
- Are there patterns of errors (e.g., a word repeatedly misspelled, a punctuation mark repeatedly used incorrectly)?

(Adapted from https://twp.duke.edu/sites/twp.duke.edu/files/file-attachments/providing_feedback%281%29.pdf)

Also refer to the slides for this lesson and watch the following video [on how to address Higher, Middle and Lower Order Concerns when peer reviewing](https://www.youtube.com/watch?v=kMWZBDUIHUA):
<https://www.youtube.com/watch?v=kMWZBDUIHUA>

Let us get practical and consider some [examples of comments you could use when peer reviewing](#). Your job is to provide constructive criticism. This means pointing out both strong points and weak points in a paper. Your feedback should be friendly, engaged, and attentive. Avoid obvious extremes: trashing the author's prose will likely do more to damage the author's ego than to improve his/her writing, while overly praising a paper gives the author little useful feedback for revising.

- Provide written feedback so that the author can refer to it when s/he revises the paper. Be specific. To improve as a writer, the author needs to know why you think one part works well or what makes another part confusing.

POOR: "This is great!"

BETTER: "Great—this example really supports what you said in the last paragraph.(Maybe it should go in the last paragraph instead of here?)"

POOR: "This doesn't make sense."

BETTER: "I don't understand this—you said X above, but now you're saying Y, which seems like the complete opposite. Do you really mean both?"

- Address local concerns—like the two examples above—with notes in the margins. At the end of the manuscript, provide a reviewer's summary, a few sentences that address the manuscript as a whole, noting what works well and what needs improvement. For example:

Hi Percy

I enjoyed reading your paper on the recent rise in popularity of organic foods. I think you have a strong thesis, and you support it with convincing evidence, especially in your paragraphs on Whole Foods and Trader Joe's. However, the paragraph on farmers' markets seems rather weak to me. I understand what you mean, but it reads more like opinion than fact. Are there any sources you could cite to make it sound more authoritative? I also wanted to mention that you often use the word "product," but I think you might mean "produce." Are those just typos? If you really mean "product," then you might want to define the term. I pointed out a few produce/product places for you above. Let me know if you have any questions about my comments. Happy revising!

Annabeth

- If you notice patterns of errors (for example, a word that is consistently misspelled, like produce/product above, or a punctuation mark that is consistently used incorrectly), then point out the pattern rather than mark all the errors.

In this course, we complete a feedback grid to guide our reviewing. Copy this grid below the first draft of each assignment. Complete some sections, and write a review paragraph below.

We could learn how to review each other's writing by considering some of the **guidelines for peer reviewing** that are given to reviewers of well-known journals such as Elsevier (Adapted from <https://www.elsevier.com/reviewers/how-to-review/checklist>). Refer to the infograph on page 8 for points to focus on when reviewing the various parts of a scientific article or thesis.

Elsevier checklist: DO

- Summarise the article in a short paragraph. This shows the editor you have read and understood the research.
- Give your main impressions of the article, including whether it is novel and interesting, whether it has a sufficient impact and adds to the knowledge base.
- Ideally when commenting, do so using short, clearly-defined paragraphs and make it easy for the editor and author to see what section you're referring to.....
- Raise your suspicions with the editor if you suspect plagiarism, fraud or have other ethical concerns, providing as much detail as possible. Visit [Elsevier's ethics page](#) or consult the [COPE guidelines](#) for more information.

The Journal of International Business Studies adds that reviewers should provide **developmental reviews**. A developmental review is specific about the problems in the manuscript and makes suggestions as to how the problems might be addressed, identifies the strengths as well as the weaknesses of the manuscript, and separates the reviewer's comments into "major" and "minor" points (HOCs and LOCs, as above), which helps signpost to the author the most important criticisms. (Adapted from http://www.palgrave-journals.com/jibs/reviewer_guidelines.html).

How to review manuscripts

Peer review, your role and responsibilities

Publishing Campus

Peer review

...is critical because it

- Improves the quality of the published paper
- Ensures previous work is acknowledged
- Determines the importance of findings
- Detects plagiarism and fraud
- Plays a central role in academic career development

...adheres to the principles that

- It is a well understood concept
- Without it there is no control in scientific communication
- Journal editors evaluate and reject certain articles prior to external peer review

Why should you review?



Editors' view: what makes a good reviewer?

- Provides a thorough and comprehensive report
- Submits the report on time
- Provides well-founded comments for authors
- Gives constructive criticism
- Demonstrates objectivity
- Provides a clear recommendation to the editor

Comments to the editor

- 1** Comment on novelty and significance
- 2** Recommend whether the manuscript is suitable for publication
- 3** Confidential comments will not be disclosed to the author(s)

Your ultimate checklist for reviewing a paper

First impressions

- Is the research original, novel and important to the field?
- Has the appropriate structure and language been used?

Abstract

- Is it really a summary?
- Does it include key findings?
- Is it an appropriate length?

Introduction

- Is it effective, clear and well organized?
- Does it really introduce and put into perspective what follows?
- Suggest changes in organization and point authors to appropriate citations.
- Be specific – don't write "the authors have done a poor job"

Methodology

- Can a colleague reproduce the experiments and get the same outcomes?
- Did the authors include proper references to previously published methodology?
- Is the description of new methodology accurate?
- Could or should the authors have included supplementary material?

Results and discussion

- Suggest improvements in the way data is shown
- Comment on general logic and on justification of interpretations and conclusions
- Comment on the number of figures, tables and schemes
- Write concisely and precisely which changes you recommend
- List separately suggested changes in style, grammar and other small changes
- Suggest additional experiments or analyses
- Make clear the need for changes/updates
- Ask yourself whether the manuscript should be published at all

Conclusion

- Comment on importance, validity and generality of conclusions
- Request toning down of unjustified claims and generalizations
- Request removal of redundancies and summaries
- The abstract, not the conclusion, summarizes the study

References, tables and figures

- Check accuracy, number and citation appropriateness
- Comment on any footnotes
- Comment on figures, their quality and readability
- Assess completeness of legends, headers and axis labels
- Check presentation consistency
- Comment on need for colour in figures

ELSEVIER

4.2 The assignment review process in this course

Reviewing is a major step in improving one's own writing and, therefore, part of the writing training in this course. Assessing writing, expressing one's views about it in writing, and making suggestions for improvement all contribute to the enhancement of one's own writing. Furthermore, it is crucial to train this skill that is central to collaborative writing, which is standard practice in publishing research. Another aspect that is also trained by reviewing other writer's work is the management of time and interpersonal relationships when reviewing and dealing with reviewers.

In this course, students review each other's assignments. **Note that participating in the review process forms part of the requirements for the course, in addition to attending classes and submitting the assignments on time.**

Open the document '**Instructions for assignments**' on ADAM. It contains full instructions for the assignments, the peer review process in this course, including the peer review schedule (who reviews whose assignment), as well as the due dates for uploading the various drafts.

During reviewing, students write a reviewer's summary paragraph of the reviewed assignment, pointing out Higher, Middle and Lower Order Concerns in the writing.

Example of a reviewer's summary of an abstract: *In this paper, the author stresses the urgent need for preventive or therapeutic interventions against Ebola [= indicating a gap]. He/she then presents a novel compound that promises to block viral replication not only in Ebola, but in a range of related viruses [= filling the gap].*

The author needs to revise the correct way of citing texts, as he/she cites inconsistently. Furthermore, there are several mistakes in the use of grammatical articles. But overall, the information flows well, in well-structured paragraphs.

Assignment 1

Assignment 1

Write a 100-200-word text in everyday spoken English in which you:

- introduce yourself, your field of study, and the topic you are researching.
- justify why the topic and field are interesting and/or significant.
- use everyday language.
- do not copy and paste a text you have written for another purpose.

Upload this assignment to your column on the Padlet page of this course:

https://sz-unibas.padlet.org/RianaPaola_szunibas/h8j5bz89xcbdw9vg